NANO LETTERS

pubs.acs.org/NanoLett Letter

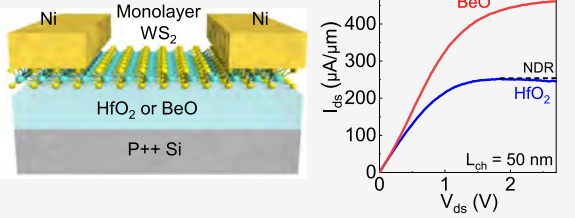
Improved Self-Heating in Short-Channel Monolayer WS_2 Transistors with High-Thermal Conductivity BeO Dielectrics

Xinhang Shi, Xuefei Li,* Qi Guo, Han Gao, Min Zeng, Yibo Han, Shiwei Yan, and Yanqing Wu*

Cite This: *Nano Lett.* 2022, 22, 7667−7673 Read Online

ACCESS | Metrics & More | Article Recommendations | Supporting Information

ABSTRACT: Two-dimensional semiconducting transition metal dichalcogenides (TMDs) enable ultimate channel length scaling of transistor technology due to their atomic-thin body nature, which also brings the challenge of a pronounced self-heating effect inside the ultrathin channel. In particular, high current density under high electric field could lead to negative differential resistance behavior due to self-heating, not only limiting the current carrying capability of the TMDs transistors but also leading to severe reliability issues. Here, we report high-performance monolayer WS_2 transistors on a high-thermal-conductivity BeO dielectric with effective suppression of the self-heating effects, eliminating the negative differential resistance behavior at high field, as observed in the case of the HfO_2 dielectric. The monolayer CVD WS_2 device on BeO with a 50 nm channel length exhibits a record-high on-state current of 325 μA/μm, transconductance (g_m) of 150 μS/μm, and a on/off ratio of 1.8×10^8 at V_{ds} = 1 V, far exceeding previous results.



KEYWORDS: *tungsten disulfide, transistor, short channel, self-heating, high thermal conductivity*

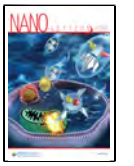
Downloaded via HUAZHONG UNIV SCIENCE & TECHNOLOGY on October 13, 2022 at 10:49:24 (UTC).
See https://pubs.acs.org/sharingguidelines for options on how to legitimately share published articles.

Two-dimensional (2D) transition-metal chalcogenides (TMDs) have attracted intensive attention in recent years, owing to their atomically thin body nature, which enables further scaling for next-generation electronic devices.[1−6] Recently, tungsten disulfide (WS_2) has attracted increasing research interest due to its larger band gap (~2 eV), smaller effective mass, and higher injection velocity among the TMDs family, all of which are vital for low-power and high-performance devices.[1,7−12] To explore the full potential of WS_2 transistors, it is of great importance to study high-field transport on short-channel devices, where heat dissipation plays a critical role, especially at high current densities. Furthermore, a recent theoretical study predicts an even lower thermal conductivity in WS_2 compared with the commonly studied MoS_2, which requires further optimizations in thermal management to prevent device performance degradation and failure.[13,14] However, most monolayer WS_2 devices reported thus far typically employ thermally oxidized 300 nm thick (or 100 nm thick) SiO_2 as the back-gate dielectric, resulting in low on-state current and transconductance (g_m), which is not only far from the desired operation region for studying power dissipation but also unsuitable for the performance boost.[1,15−17] Short channel devices in advanced technology nodes typically adopt a high-κ dielectric for a smaller equivalent oxide thickness (EOT) as required by transistor scaling.

Although hafnium dioxide (HfO_2) is the most widely used high-κ dielectric, the thermal conductivity is quite small (~23 W/K·m).[18] Compared with HfO_2, beryllium oxide (BeO) has a much higher thermal conductivity (~330 W/K·m), which is beneficial for alleviating the self-heating effect and improving device performance and reliability.[18] On the other hand, it also exhibits a larger bandgap (E_g = 10.6 eV) with a decent dielectric constant (~8.7), which helps reduce the gate leakage current at higher electric fields.[19] Meanwhile, it can also be grown by atomic layer deposition (ALD) with precise thickness control on a wafer scale.[19] Previously, BeO was used as a gate dielectric in III−V and Si field-effect transistors (FETs), exhibiting excellent interface quality.[20] Nevertheless, the advantageous thermal conductivity of BeO in atomic thin TMDs-based transistors has yet to be explored.

In this work, we demonstrate high-performance n-type monolayer CVD WS_2 FETs with a high-quality BeO dielectric. The average carrier mobility of monolayer WS_2 on the BeO dielectric is 62% higher than that on the HfO_2 dielectric. The monolayer WS_2 transistors on BeO dielectrics exhibit a record-high on-state current of 325 μA/μm at a channel length (L_{ch}) of 50 nm at a source-drain voltage (V_{ds}) of 1 V, which increases to 461 μA/μm at V_{ds} = 2.75 V without negative differential resistance behavior. The gate oxide quality of WS_2 devices is also evaluated by low-frequency noise, and the oxide trap

Received: July 22, 2022
Revised: September 1, 2022
Published: September 9, 2022



ACS Publications © 2022 American Chemical Society

https://doi.org/10.1021/acs.nanolett.2c02901
Nano Lett. 2022, 22, 7667−7673

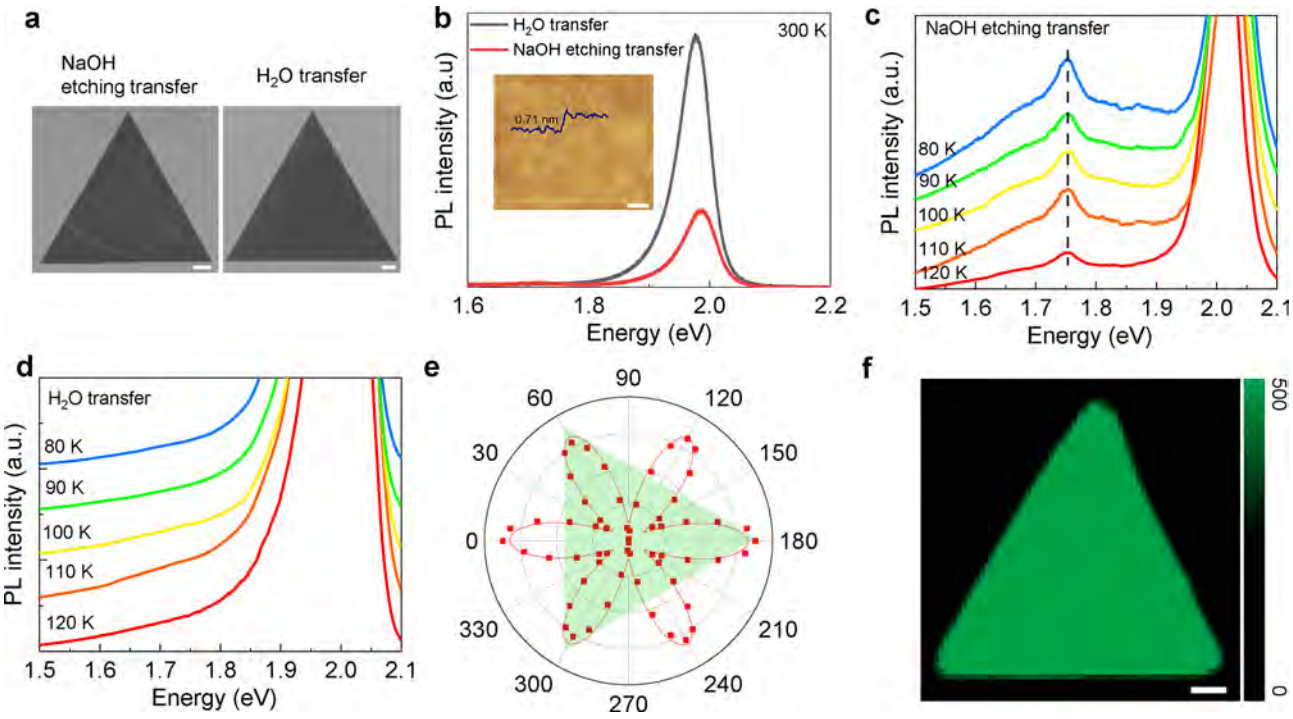


Figure 1. (a) SEM images of monolayer WS_2 on SiO_2 using NaOH (left panel) or water (right panel) transfer. The scale bar is 2 μm. (b) The PL spectra of monolayer WS_2 on SiO_2 using NaOH or water transfer at room temperature. The inset is the AFM image of monolayer WS_2 on SiO_2/Si substrates using water transfer. The scale bar is 2 μm. Temperature-dependent PL spectra of monolayer CVD WS_2 with (c) NaOH etching transfer and (d) water transfer. (e) Polarization angle-dependent SHG intensity of monolayer WS_2 flake on SiO_2 using water transfer with 6-fold rotational symmetry. (f) SHG intensity mapping of a typical triangular WS_2 flake on SiO_2 using water transfer. The scale bar is 2 μm.

density (N_{ot}) of the WS_2 device based on BeO is 2.4 times lower than that on HfO_2.

Triangular high-quality monolayer WS_2 single crystals on sapphire substrates can be synthesized using a single-step CVD process with WO_2 powder as the tungsten precursor. The size of a high-quality equilateral triangle monolayer WS_2 can reach ~80 μm in lateral length (Figure S1). Moreover, to meet the demands of characterization and device fabrication, monolayer WS_2 is transferred from sapphire substrates onto target dielectric substrates (such as SiO_2, HfO_2, and BeO) using DI water or the NaOH etching transfer method.[12] The details of the transfer process can be found in Figure S2. Scanning electron microscopy (SEM) images of the transferred WS_2 film are shown in Figure 1a. It is clear that the WS_2 films using the NaOH etching transfer process exhibit wrinkles and cracks, indicating film quality degradation.[21] Moreover, Na^+ ion diffusion can cause threshold voltage shift in WS_2 device, resulting in low reliability. WS_2 films using deionized (DI) water transfer can take advantage of the different natures of hydrophobic WS_2, and hydrophilic c-plan sapphire retains the original sample morphology and quality.[22] Figure 1b compares the typical photoluminescence (PL) intensity of WS_2 films using the two different transfer methods. Compared with the NaOH etching method, the PL intensity of the WS_2 films using the DI water transfer method is higher, indicating that WS_2 is well protected from damage with high quality.[21,23] The inset of Figure 1b shows an atomic force microscopy (AFM) image of the water-assisted transferred WS_2 film on the SiO_2/Si substrate. Obviously, there are no observable cracks, wrinkles, or holes in the WS_2 film. The height profile indicates that the thickness of single monolayer WS_2 is ~0.71 nm, which is in good agreement with the theoretical physical thickness.[7] To further confirm the crystal quality of the transferred WS_2 film, we also investigated the low-temperature PL of the WS_2 film, as shown in Figure 1c. It is clear that the PL spectra always exhibit a side peak centered at 1.75 eV, and the peak intensity increases by decreasing from 120 to 80 K. This is mainly attributed to lattice defects and bound excitons in the low-quality WS_2 material.[10,23−25] As expected, the absence of a side peak at low temperature in Figure 1d indicates the high quality of the water-assisted transferred WS_2 film without any damage.[10,25] The Raman spectra in Figure S3 also confirm the high quality of the water-assisted transferred WS_2. Second-harmonic generation (SHG) is further applied to characterize the crystalline quality of WS_2 film with water transfer because monolayer WS_2 lacks inversion symmetry and has strong optical SHG.[26] Based on this feature, an optical experiment setup is built to characterize SHG in monolayer WS_2 on SiO_2 (Figure S4). The polarization-dependent SHG intensity pattern is measured by changing the polarization direction of the excitation laser with a power of 2 mW. The angle of linear polarization is changed by the rotation of the half-wave plate before the WS_2 films. The 6-fold rotationally symmetrical shape of the SHG intensity is illustrated in Figure 1e, which indicates the typical hexagonal symmetric feature of the monolayer WS_2 crystal.[7] Furthermore, SHG mapping of the monolayer WS_2 in Figure 1f demonstrates good homogeneity of the single crystal without any grain boundaries and wrinkles.[5]

To characterize the scaling properties of monolayer WS_2 with water transfer, back-gated FETs with channel lengths from 3 μm down to 50 nm were fabricated on 10 nm thick HfO_2 or 8 nm thick BeO dielectric substrates with Si back gate (Figures S6−S8 provide the detailed characterization of BeO

https://doi.org/10.1021/acs.nanolett.2c02901
Nano Lett. 2022, 22, 7667−7673

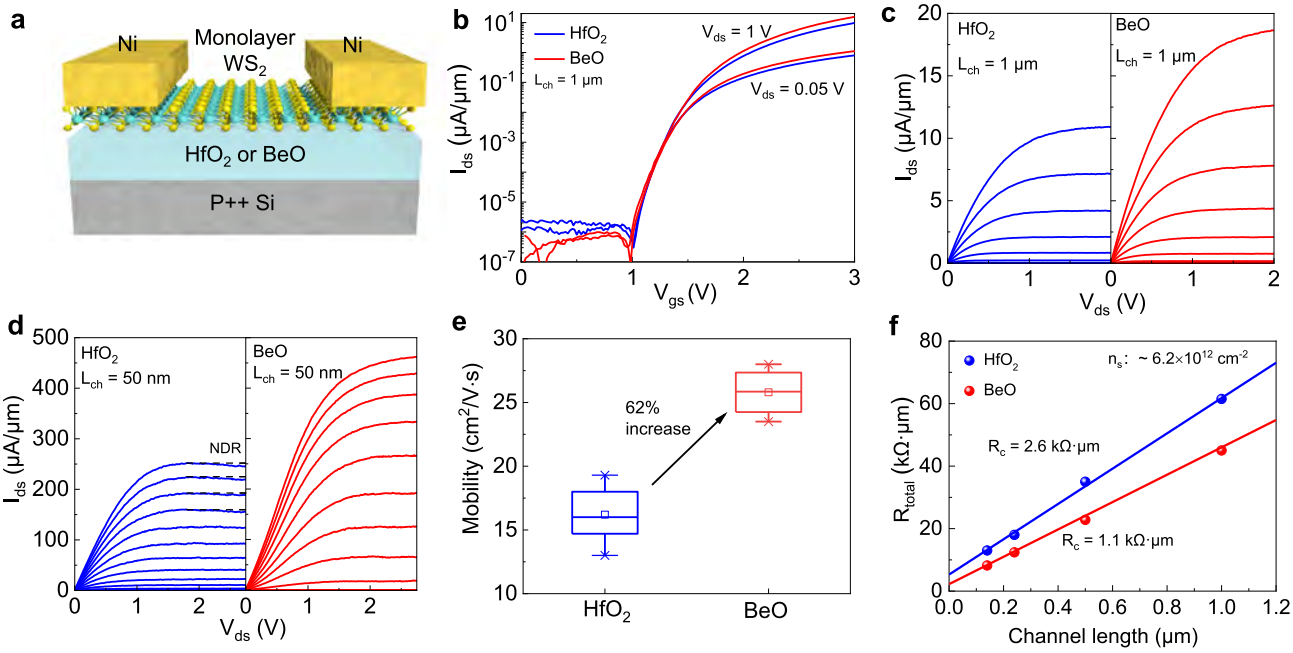


Figure 2. (a) Schematic of the monolayer WS_2 back-gate FET fabricated in this work. The gate dielectric is 10 nm HfO_2 or 8 nm BeO. The S/D contact metal is Ni (20 nm)/Au (40 nm). (b) $I_{ds}-V_{gs}$ transfer curves of monolayer WS_2 transistors with 10 nm HfO_2 and 8 nm BeO dielectrics. Both the channel length and channel width are 1 μm. (c) The corresponding $I_{ds}-V_{ds}$ output characteristics of the monolayer WS_2 transistors. V_{gs} changes from 0 to 3 V. (d) The $I_{ds}-V_{ds}$ output characteristics of the monolayer WS_2 transistors with a channel length of 50 nm. V_{gs} changes from −0.5 to 4 V. (e) Statistical distribution of field-effect mobility of water-assisted transferred monolayer WS_2 on HfO_2 or BeO dielectrics. (f) TLM resistances of monolayer WS_2 transistors with HfO_2 and BeO dielectrics. R_c is extracted to be 2.6 kΩ·μm and 1.1 kΩ·μm at $n_s \sim 6.2 \times 10^{12}$ cm^{-2} for HfO_2 and BeO, respectively.

dielectrics). Figure 2a illustrates the schematic of the monolayer WS_2 transistor in this work. The gate stack includes HfO_2 or BeO as the gate dielectric and monolayer CVD WS_2 as the semiconducting channels. Figure S5 shows a scanning electron microscopy (SEM) image of a fabricated transmission line method (TLM) structure with a channel length of 50 nm in this work. To compare the effect of the dielectric substrate on the electrical properties, the transfer characteristics of the monolayer WS_2 devices with HfO_2 and BeO dielectrics with a channel length of 1 μm are shown in Figure 2b. Both devices show n-type transport behaviors, similar positive threshold voltages, and high on/off ratios of 10^8. The field-effect mobility (μ_{FE}) of the monolayer WS_2 devices with HfO_2 and BeO dielectrics extracted by peak transconductance is 16 and 26.7 $cm^2/V{\cdot}s$, respectively.[5] The corresponding output characteristics are plotted in Figure 2c. As expected, the drain current of WS_2 transistors based on BeO dielectrics is approximately 65% times larger than that on HfO_2 dielectrics at similar carrier densities, up to 18.6 $\mu A/\mu m$ at V_{ds} = 2 V. Apart from mobility, on-state current density (I_{on}), especially in short-channel devices, is also a practically more important parameter for evaluating the potential of 2D semiconductors.[1] The output characteristics of monolayer WS_2 with HfO_2 and BeO dielectrics with a 50 nm channel length at room temperature are shown in Figure 2d. The drain current I_{ds} is 222 $\mu A/\mu m$ at V_{ds} = 1 V for the device based on the HfO_2 dielectric and reaches maxima at V_{ds} = 2 V. When V_{ds} is larger than 2 V, the drain current decreases monotonously for I_{ds} > 150 $\mu A/\mu m$, exhibiting obvious negative differential resistance (NDR) behavior on HfO_2 due to self-heating at high V_{ds}, which is consistent with a previous report on CVD-grown monolayer WS_2 FETs.[27] On the other hand, the WS_2 device with a BeO dielectric shows well-behaved drain saturation and achieves a record high current density of 325 $\mu A/\mu m$ at V_{ds} = 1 V. Furthermore, this short-channel device experiences high-field current saturation for $V_{ds} \sim 2.0$ V, with a record high room temperature current density >400 $\mu A/\mu m$. Importantly, no NDR effect is observed in the WS_2 device based on the BeO dielectric, with a significant 84% improvement in drain current over that on the HfO_2 dielectric, indicating efficient heat dissipation using the BeO dielectric due to its high thermal conductivity.[1,17,18,27] Note that the study on the effect of WS_2 thickness on the thermal conductivity of WS_2 has been limited, which needs further investigation, including experiments and theory. The saturation velocity (v_{sat}) is extracted from the saturation output characteristics, and v_{sat} values of 1.0×10^6 cm/s and 1.8×10^6 cm/s are obtained for HfO_2/WS_2 and BeO/WS_2, respectively (Figure S9).[27] This indicates that using a high-thermal-conductivity gate dielectric is beneficial to improve v_{sat}. Figure 2e compares the statistical analysis of the mobility of WS_2 devices using HfO_2 and BeO dielectrics. The average μ_{FE} is 16 $cm^2/V{\cdot}s$ for the devices based on the HfO_2 dielectric and 26 $cm^2/V{\cdot}s$ for the devices based on the BeO dielectric. The contact resistance (R_c), a key figure of merit in a transistor, has also been extracted by TLM, as shown in Figure 2f. R_c values of 2.6 kΩ·μm and 1.1 kΩ·μm at $n_s \sim 6.2 \times 10^{12}$ cm^{-2} are obtained for monolayer WS_2 devices on HfO_2 and BeO dielectrics, respectively. The decrease in R_c can be attributed to the higher mobility for BeO/WS_2 devices.

To further evaluate the performance of monolayer WS_2 transistors with HfO_2 and BeO dielectrics, low-frequency ($1/f$) noise characterization is performed to characterize the noise behavior and the quality of their interfaces.[28] The $1/f$ noise was measured in the linear region at V_{ds} = 0.2 V at 300 K. Normalized noise power spectra ($W^*L^*S_{id}/I_{ds}^2$) of the WS_2 devices with HfO_2 and BeO dielectrics between 1 Hz and 1

https://doi.org/10.1021/acs.nanolett.2c02901
Nano Lett. 2022, 22, 7667−7673

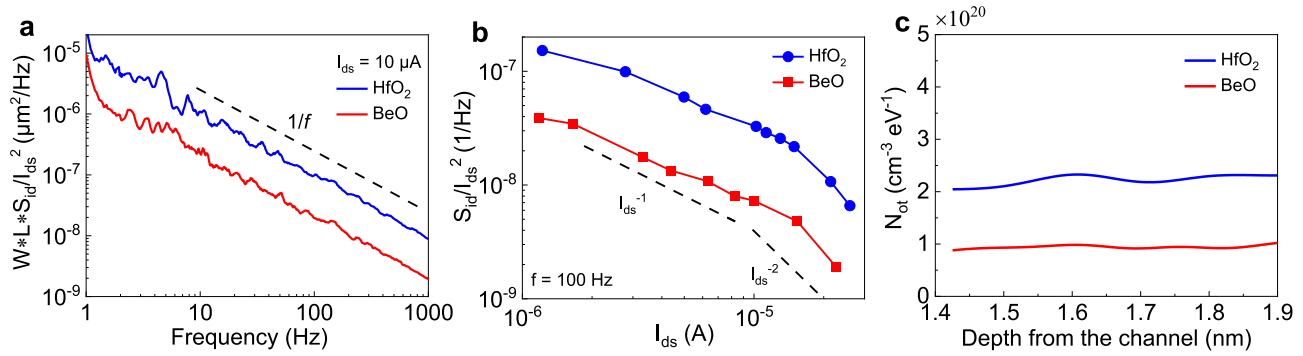


Figure 3. (a) Normalized noise spectral density as a function of frequency for HfO_2 and BeO at similar drain currents. (b) Current spectral density S_{id}/I_{ds}^2 d at f = 100 Hz versus drain current I_{ds} for the monolayer WS_2 devices with HfO_2 and BeO at room temperature. (c) Oxide trap density distribution in dielectrics extracted from the low-frequency noise spectra.

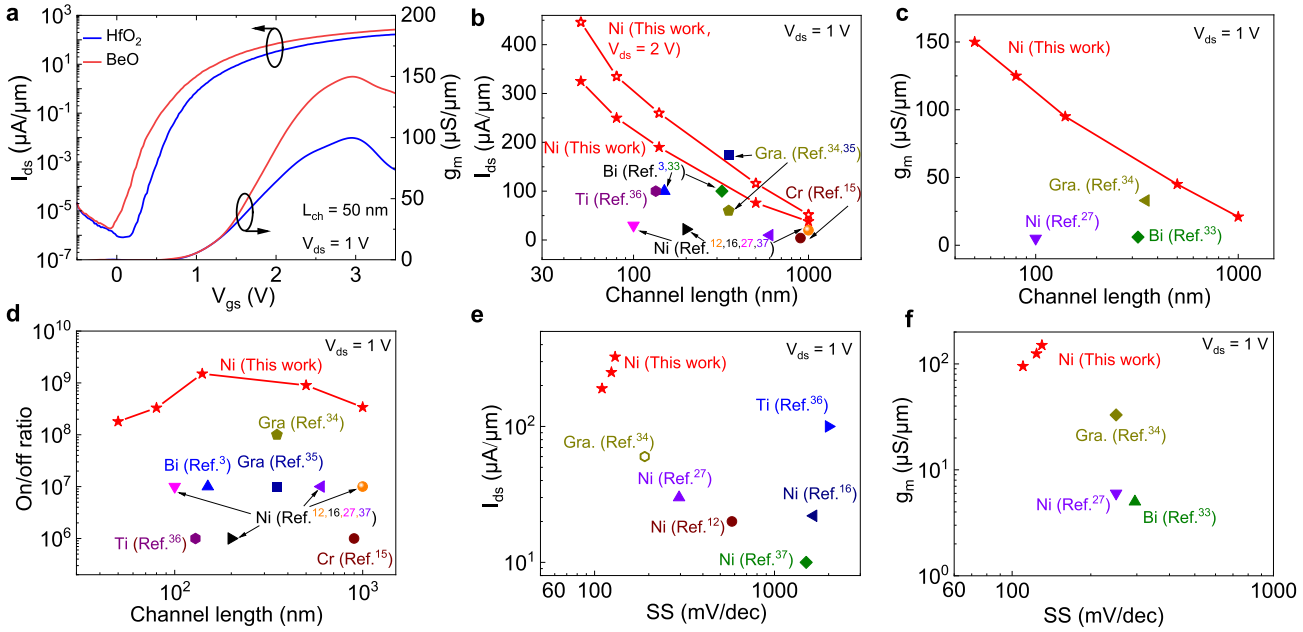


Figure 4. (a) I_{ds}−V_{gs} transfer curves and g_m of the 50 nm L_{ch} monolayer WS_2 transistors with HfO_2 and BeO dielectrics. (b−d) Benchmark of I_{ds}, g_m, and on/off ratio versus channel length for monolayer CVD WS_2 transistors. In this work, the maximum I_{ds} are extracted at V_{gs} = 4 V. (e,f) Benchmark of I_{ds} and g_m versus SS for monolayer CVD WS_2 transistors. All the data from the literature are extracted at V_{ds} = 1 V at room temperature. Our WS_2 devices based on BeO have the highest I_{ds}, g_m, and on/off ratio on CVD-1L-WS_2.

kHz at a fixed drain current of 10 μA are shown in Figure 3a. The noise spectra follow the typical 1/f dependence. It is clear that the noise magnitude of BeO/WS_2 is lower than that of HfO_2/WS_2. To further investigate the noise mechanism of 1/f noise, S_{id}/I_{ds}^2 at a frequency of 100 Hz as a function of the drain current I_{ds} of the same two devices is plotted in Figure 3b. It is observed that S_{id}/I_{ds}^2 has the same dependence on I_{ds} for both devices, indicating the same noise mechanism. When the drain current is below 10 μA, S_{id}/I_{ds}^2 is proportional to 1/I_{ds}. Above 10 μA, S_{id}/I_{ds}^2 follows the $1/I_{ds}^2$ law. This shows that the carrier number fluctuation and the correlated mobility fluctuation are the main sources of 1/f noise.[29−31] We can also estimate the oxide trap density using the input-referred voltage noise spectral density at the threshold voltage, and the profile for both dielectrics is shown in Figure 3c.[17,32] The average N_{ot} is 9.0 × 10^{19} cm^{-3} eV^{-1} for the BeO/WS_2 device, which is much lower than the 2.2 × 10^{20} cm^{-3} eV^{-1} for the HfO_2/WS_2 device. This result indicates a lower interface trap density of BeO/WS_2 with high interface quality, in agreement with our observation from the electrical results. The low-frequency noise at 10 Hz at room temperature is as low as 8.4 × 10^{-8} μm^2 Hz^{-1}, which is approximately 289 times the reduction from previous reports (Figure S10 in Supporting Information).[17,31]

Figure 4a shows the transfer characteristics and transconductance g_m of the monolayer WS_2 device with HfO_2 and BeO dielectrics with a channel length of 50 nm at room temperature. The BeO device exhibits a high on/off ratio of 1.8 × 10^8 at V_{ds} of 1 V due to the relatively large bandgap of monolayer WS_2 (~2.0 eV) and atomically thin channel. Compared with ultrathin silicon on oxide, atomically thin channel can effectively reduce the natural length, resulting in negligible short channel effect. A large on/off ratio represents a better on-state current and lower static power consumption, which is essential for low power applications. Different from the commonly reported n-type MoS_2 transistor, the threshold voltage of the short-channel WS_2 device is positive, which is of great benefit to fabricating an enhancement-mode transistor.[3,5] As expected, the gate leakage of BeO/WS_2 devices is smaller than that of HfO_2/WS_2, indicating the excellent dielectric

https://doi.org/10.1021/acs.nanolett.2c02901
Nano Lett. 2022, 22, 7667−7673

properties of BeO (Figure S11). Figure S12 shows the hysteresis curves of BeO/WS_2 device at room temperature. The small hysteresis indicates a less defective gate insulator. The maximum g_m of 150 $\mu S/\mu m$ is achieved for the BeO/WS_2 device at L_{ch} of 50 nm and V_{ds} = 1 V, 50% higher than that of the HfO_2/WS_2 device. Integrating ultrathin BeO with an insulator having higher permittivity to form a gate stack can further reduce EOT and improve transconductance. Figure 4b shows the benchmark of I_{on} as a function of L_{ch} at V_{ds} = 1 V in comparison with previously reported monolayer WS_2 transistors (see Table 1 in Supporting Information for details).[3,12,15,27,33−37] The on-state current of 325 $\mu A/\mu m$ is the largest reported among those from the literature on monolayer CVD WS_2. Note that I_{on} exhibits a monotonic increase as the channel length decreases, which is beneficial for further device scaling. The transconductance g_m of monolayer WS_2 transistors with different channel lengths are also compared, as shown in Figure 4c, with fixed V_{ds} at 1 V for all data.[3,27,33,34] Compared with monolayer WS_2 transistors in previous reports, WS_2 with a 50 nm channel length in this work exhibits the highest g_m of 150 $\mu S/\mu m$ at V_{ds} = 1 V. Figure 4d summarizes the on/off ratio of monolayer WS_2 devices at different channel lengths. We achieve the record high on/off ratio, which is the highest among the reported TMDs semiconductor transistors for the same L_{ch}.[3,5,9] For long-channel devices, the I_{on}/I_{off} ratios are more than 1×10^9. The on/off ratio increases with the channel length scaling due to the maximum drain current increase. Figure 4e,f plots the I_{on} and g_m versus subthreshold slope (SS) for monolayer CVD WS_2 FETs at V_{ds} = 1 V, respectively.[3,12,15,27,33,34,36,37] Our devices show the best overall performance with the highest I_{on}, g_m, and on/off ratio as well as the lowest SS. These benchmark results demonstrate the potential of monolayer WS_2 with a BeO dielectric for future high-performance electronic devices with minimal self-heating effect.

In summary, we have demonstrated high-performance monolayer CVD WS_2 transistors on a high-thermal-conductivity BeO dielectric for the first time, which effectively facilitates heat dissipation and suppresses the self-heating effect. Notably, the 50 nm short channel device shows a high I_{on}/I_{off} ratio of 1.8×10^8, g_m of 150 $\mu S/\mu m$, and I_{on} of 325 $\mu A/\mu m$ at V_{ds} = 1 V, which are record values among all reported monolayer CVD WS_2 transistors. The promising results of this work provide a scalable and reliable method to achieve high-performance TMDs transistors for future electronics.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.nanolett.2c02901.

Additional results and details on CVD growth and transfer monolayer WS_2, device fabrication and characterization, Raman and SHG characterization of monolayer WS_2, XPS and SEM measurement, MOS capacitor results for BeO, gate leakage and hysteresis curves of WS_2 devices, and noise level comparison (PDF)

■ AUTHOR INFORMATION

Corresponding Authors

Xuefei Li − *Wuhan National High Magnetic Field Center and School of Optical and Electronic Information, Huazhong University of Science and Technology, Wuhan 430074, China;* orcid.org/0000-0001-6406-6461; Email: xfli@hust.edu.cn

Yanqing Wu − *Wuhan National High Magnetic Field Center, Huazhong University of Science and Technology, Wuhan 430074, China; School of Integrated Circuits, Peking University, Beijing 100871, China;* orcid.org/0000-0003-2578-5214; Email: yqwu@pku.edu.cn

Authors

Xinhang Shi − *Wuhan National High Magnetic Field Center and School of Optical and Electronic Information, Huazhong University of Science and Technology, Wuhan 430074, China*

Qi Guo − *Wuhan National High Magnetic Field Center and School of Optical and Electronic Information, Huazhong University of Science and Technology, Wuhan 430074, China*

Han Gao − *Wuhan National High Magnetic Field Center and School of Optical and Electronic Information, Huazhong University of Science and Technology, Wuhan 430074, China*

Min Zeng − *Wuhan National High Magnetic Field Center and School of Optical and Electronic Information, Huazhong University of Science and Technology, Wuhan 430074, China*

Yibo Han − *Wuhan National High Magnetic Field Center and School of Optical and Electronic Information, Huazhong University of Science and Technology, Wuhan 430074, China;* orcid.org/0000-0003-1296-5718

Shiwei Yan − *Wuhan National High Magnetic Field Center and School of Optical and Electronic Information, Huazhong University of Science and Technology, Wuhan 430074, China*

Complete contact information is available at: https://pubs.acs.org/10.1021/acs.nanolett.2c02901

Author Contributions

X.L. and Y.W. conceived the project. X.S. fabricated these devices. X.S, H.G, and Q.G. performed the optical and AFM measurements. M.Z. and S.Y. performed the capacitor measurements. X.L, X.S., and Y.W. analyzed the data and wrote the manuscript. All authors commented on the manuscript.

Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

This work was supported by the National Key Research and Development Program of China (2021YFA1202903), the National Natural Science Foundation of China (Grants 91964106, 61874162, 62090034, and 61927901), the Fundamental Research Funds for the Central Universities (Grant 2018KFYYXJJ069), the Strategic Priority Research Program of Chinese Academy of Sciences (Grant XDB30000000), and the 111 Project (B18001).

■ REFERENCES

(1) Liu, Y.; Duan, X.; Shin, H.-J.; Park, S.; Huang, Y.; Duan, X. Promises and prospects of two-dimensional transistors. *Nature* **2021**, *591* (7848), 43−53.

(2) Desai, S. B.; Madhvapathy, S. R.; Sachid, A. B.; Llinas, J. P.; Wang, Q.; Ahn, G. H.; Pitner, G.; Kim, M. J.; Bokor, J.; Hu, C.; Wong, H. S. P.; Javey, A. MoS_2 transistors with 1-nanometer gate lengths. *Science* **2016**, *354* (6308), 99.

(3) Shen, P.-C.; Su, C.; Lin, Y.; Chou, A.-S.; Cheng, C.-C.; Park, J.-H.; Chiu, M.-H.; Lu, A.-Y.; Tang, H.-L.; Tavakoli, M. M.; Pitner, G.; Ji, X.; Cai, Z.; Mao, N.; Wang, J.; Tung, V.; Li, J.; Bokor, J.; Zettl, A.; Wu, C.-I.; Palacios, T.; Li, L.-J.; Kong, J. Ultralow contact resistance

between semimetal and monolayer semiconductors. *Nature* **2021**, *593* (7858), 211−217.
(4) Hong, S.; Zagni, N.; Choo, S.; Liu, N.; Baek, S.; Bala, A.; Yoo, H.; Kang, B. H.; Kim, H. J.; Yun, H. J.; Alam, M. A.; Kim, S. Highly sensitive active pixel image sensor array driven by large-area bilayer MoS_2 transistor circuitry. *Nat. Commun.* **2021**, *12* (1), 3559.
(5) Liu, L.; Li, T.; Ma, L.; Li, W.; Gao, S.; Sun, W.; Dong, R.; Zou, X.; Fan, D.; Shao, L.; Gu, C.; Dai, N.; Yu, Z.; Chen, X.; Tu, X.; Nie, Y.; Wang, P.; Wang, J.; Shi, Y.; Wang, X. Uniform nucleation and epitaxy of bilayer molybdenum disulfide on sapphire. *Nature* **2022**, *605* (7908), 69−75.
(6) Du, Y.; Yang, L.; Liu, H.; Ye, P. D. Contact research strategy for emerging molybdenum disulfide and other two-dimensional field-effect transistors. *APL Materials* **2014**, *2* (9), 092510.
(7) Lan, C.; Li, C.; Ho, J. C.; Liu, Y. 2D WS_2: From Vapor Phase Synthesis to Device Applications. *Advanced Electronic Materials* **2021**, *7* (7), 2000688.
(8) Jin, Z.; Li, X.; Mullen, J. T.; Kim, K. W. Intrinsic transport properties of electrons and holes in monolayer transition-metal dichalcogenides. *Phys. Rev. B* **2014**, *90* (4), 045422.
(9) Cheng, C. C.; Chung, Y. Y.; Li, U. Y.; Lin, C. T.; Li, C. F.; Chen, J. H.; Lai, T. Y.; Li, K. S.; Shieh, J. M.; Su, S. K.; Chiang, H. L.; Chen, T. C.; Li, L. J.; Wong, H. S. P.; Chien, C. H. In *First demonstration of 40-nm channel length top-gate WS_2 pFET using channel area-selective CVD growth directly on SiOx/Si substrate*; 2019 Symposium on VLSI Technology, 9−14 June 2019; 2019; pp T244−T245.
(10) Wang, J.; Xu, X.; Cheng, T.; Gu, L.; Qiao, R.; Liang, Z.; Ding, D.; Hong, H.; Zheng, P.; Zhang, Z.; Zhang, Z.; Zhang, S.; Cui, G.; Chang, C.; Huang, C.; Qi, J.; Liang, J.; Liu, C.; Zuo, Y.; Xue, G.; Fang, X.; Tian, J.; Wu, M.; Guo, Y.; Yao, Z.; Jiao, Q.; Liu, L.; Gao, P.; Li, Q.; Yang, R.; Zhang, G.; Tang, Z.; Yu, D.; Wang, E.; Lu, J.; Zhao, Y.; Wu, S.; Ding, F.; Liu, K. Dual-coupling-guided epitaxial growth of wafer-scale single-crystal WS_2 monolayer on vicinal a-plane sapphire. *Nat. Nanotechnol.* **2022**, *17* (1), 33−38.
(11) Zhang, Z.; Chen, P.; Yang, X.; Liu, Y.; Ma, H.; Li, J.; Zhao, B.; Luo, J.; Duan, X.; Duan, X. Ultrafast growth of large single crystals of monolayer WS_2 and WSe_2. *National Science Review* **2020**, *7* (4), 737−744.
(12) Chubarov, M.; Choudhury, T. H.; Hickey, D. R.; Bachu, S.; Zhang, T.; Sebastian, A.; Bansal, A.; Zhu, H.; Trainor, N.; Das, S.; Terrones, M.; Alem, N.; Redwing, J. M. Wafer-scale epitaxial growth of unidirectional WS_2 monolayers on sapphire. *ACS Nano* **2021**, *15* (2), 2532−2541.
(13) Peng, B.; Zhang, H.; Shao, H.; Xu, Y.; Zhang, X.; Zhu, H. Thermal conductivity of monolayer MoS_2, $MoSe_2$, and WS_2: interplay of mass effect, interatomic bonding and anharmonicity. *RSC Adv.* **2016**, *6* (7), 5767−5773.
(14) Yalon, E.; McClellan, C. J.; Smithe, K. K. H.; Muñoz Rojo, M.; Xu, R. L.; Suryavanshi, S. V.; Gabourie, A. J.; Neumann, C. M.; Xiong, F.; Farimani, A. B.; Pop, E. Energy dissipation in monolayer MoS_2 electronics. *Nano Lett.* **2017**, *17* (6), 3429−3433.
(15) Li, S.; Wang, S.; Tang, D.-M.; Zhao, W.; Xu, H.; Chu, L.; Bando, Y.; Golberg, D.; Eda, G. Halide-assisted atmospheric pressure growth of large WSe_2 and WS_2 monolayer crystals. *Applied Materials Today* **2015**, *1* (1), 60−66.
(16) Brien, K. P. O.; Dorow, C. J.; Penumatcha, A.; Maxey, K.; Lee, S.; Naylor, C. H.; Hsiao, A.; Holybee, B.; Rogan, C.; Adams, D.; Tronic, T.; Ma, S.; Oni, A.; Gupta, A. S.; Bristol, R.; Clendenning, S.; Metz, M.; Avci, U. In *Advancing 2D monolayer CMOS through contact, channel and interface engineering*; 2021 IEEE International Electron Devices Meeting (IEDM), 11−16 Dec. 2021; 2021; pp 7.1.1−7.1.4.
(17) Li, X.; Xiong, X.; Li, T.; Li, S.; Zhang, Z.; Wu, Y. Effect of dielectric interface on the performance of MoS_2 transistors. *ACS Appl. Mater. Interfaces* **2017**, *9* (51), 44602−44608.
(18) Slack, G. A.; Austerman, S. B. Thermal conductivity of BeO single crystals. *J. Appl. Phys.* **1971**, *42* (12), 4713−4717.
(19) Yum, J. H.; Bersuker, G.; Akyol, T.; Ferrer, D. A.; Lei, M.; Park, K. W.; Hudnall, T. W.; Downer, M. C.; Bielawski, C. W.; Yu, E. T.; Price, J.; Lee, J. C.; Banerjee, S. K. Epitaxial ALD BeO: efficient oxygen diffusion barrier for EOT scaling and reliability improvement. *IEEE Trans. Electron Devices* **2011**, *58* (12), 4384−4392.
(20) Yum, J. H.; Akyol, T.; Lei, M.; Hudnall, T.; Bersuker, G.; Downer, M.; Bielawski, C. W.; Lee, J. C.; Banerjee, S. K. Atomic layer deposited beryllium oxide: effective passivation layer for III-V metal/oxide/semiconductor devices. *J. Appl. Phys.* **2011**, *109* (6), 064101.
(21) Watson, A. J.; Lu, W.; Guimarães, M. H. D.; Stöhr, M. Transfer of large-scale two-dimensional semiconductors: challenges and developments. *2D Materials* **2021**, *8* (3), 032001.
(22) Gurarslan, A.; Yu, Y.; Su, L.; Yu, Y.; Suarez, F.; Yao, S.; Zhu, Y.; Ozturk, M.; Zhang, Y.; Cao, L. Surface-energy-assisted perfect transfer of centimeter-scale monolayer and few-layer MoS_2 films onto arbitrary substrates. *ACS Nano* **2014**, *8* (11), 11522−11528.
(23) Rosenberger, M. R.; Chuang, H.-J.; McCreary, K. M.; Li, C. H.; Jonker, B. T. Electrical characterization of discrete defects and impact of defect density on photoluminescence in monolayer WS_2. *ACS Nano* **2018**, *12* (2), 1793−1800.
(24) Jeong, B. G.; Lee, C.; Kim, S. H.; Yun, S. J.; Kim, D. H.; Lee, J.; Lee, D.; Kim, K. K.; Lim, S. C.; Jeong, M. S. Correlation of defect-induced photoluminescence and Raman scattering in monolayer WS_2. *J. Phys. Chem. C* **2022**, *126* (16), 7177−7183.
(25) Carozo, V.; Wang, Y.; Fujisawa, K.; Carvalho, B. R.; McCreary, A.; Feng, S.; Lin, Z.; Zhou, C.; Perea-López, N.; Elías, A. L.; Kabius, B.; Crespi, V. H.; Terrones, M. Optical identification of sulfur vacancies: Bound excitons at the edges of monolayer tungsten disulfide. *Science Advances* **2017**, *3* (4), No. e1602813.
(26) Zeng, Z.; Sun, X.; Zhang, D.; Zheng, W.; Fan, X.; He, M.; Xu, T.; Sun, L.; Wang, X.; Pan, A. Controlled vapor growth and nonlinear optical applications of large-area 3R phase WS_2 and WSe_2 atomic layers. *Adv. Funct. Mater.* **2019**, *29* (11), 1806874.
(27) Sebastian, A.; Pendurthi, R.; Choudhury, T. H.; Redwing, J. M.; Das, S. Benchmarking monolayer MoS_2 and WS_2 field-effect transistors. *Nat. Commun.* **2021**, *12* (1), 693.
(28) Vandamme, L. K. J. Bulk and surface 1/*f* noise. *IEEE Trans. Electron Devices* **1989**, *36* (5), 987−992.
(29) Li, X.; Yang, L.; Si, M.; Li, S.; Huang, M.; Ye, P.; Wu, Y. Performance potential and limit of MoS_2 transistors. *Adv. Mater.* **2015**, *27* (9), 1547−1552.
(30) Vandamme, L. K. J.; Hooge, F. N. What do we certainly know about 1/*f* noise in MOSFETs? *IEEE Trans. Electron Devices* **2008**, *55* (11), 3070−3085.
(31) Joo, M.-K.; Yun, Y.; Yun, S.; Lee, Y. H.; Suh, D. Strong coulomb scattering effects on low frequency noise in monolayer WS_2 field-effect transistors. *Appllied Physics Letters* **2016**, *109* (15), 153102.
(32) Scarpino, M.; Gupta, S.; Lin, D.; Alian, A.; Crupi, F.; Collaert, N.; Thean, A.; Simoen, E. Border traps in InGaAs nMOSFETs assessed by low-frequency Noise. *IEEE Electron Device Lett.* **2014**, *35* (7), 720−722.
(33) Jin, L.; Koester, S. J. High-performance dual-gated single-layer WS_2 MOSFETs with Bi contacts. *IEEE Electron Device Lett.* **2022**, *43* (4), 639−642.
(34) Yeh, C.-H.; Liang, Z.-Y.; Lin, Y.-C.; Ma, C.-H.; Chu, Y.-H.; Suenaga, K.; Chiu, P.-W. Scalable T-gate aligned Gr-WS_2-Gr radio-frequency field-effect transistors. *ACS Applied Electronic Materials* **2020**, *2* (12), 3898−3905.
(35) Yeh, C.; Cao, W.; Pal, A.; Parto, K.; Banerjee, K. In *Area-selective-CVD technology enabled top-gated and scalable 2D-hetero-junction transistors with dynamically tunable Schottky barrier*; 2019 IEEE International Electron Devices Meeting (IEDM), 7−11 Dec. 2019; 2019; pp 23.4.1−23.4.4.
(36) Smets, Q.; Schram, T.; Verreck, D.; Cott, D.; Groven, B.; Ahmed, Z.; Kaczer, B.; Mitard, J.; Wu, X.; Kundu, S.; Mertens, H.; Radisic, D.; Thiam, A.; Li, W.; Dupuy, E.; Tao, Z.; Vandersmissen, K.; Maurice, T.; Lin, D.; Morin, P.; Asselberghs, I.; Radu, I. In *Scaling of double-gated WS_2 FETs to sub-5 nm physical gate length fabricated in a 300 mm FAB*; 2021 IEEE International Electron Devices Meeting (IEDM), 11−16 Dec. 2021; 2021; pp 34.2.1−34.2.4.
(37) Dorow, C. J.; O'Brien, K. P.; Naylor, C. H.; Lee, S.; Penumatcha, A.; Hsiao, A.; Tronic, T.; Christenson, M.; Maxey, K.;

https://doi.org/10.1021/acs.nanolett.2c02901
Nano Lett. 2022, 22, 7667−7673

Zhu, H.; Oni, A.; Alaan, U. S.; Gosavi, T. A.; Gupta, A. S.; Bristol, R.; Clendenning, S.; Metz, M.; Avci, U. E. In *Advancing monolayer 2D NMOS and PMOS transistor integration from growth to van der Waals interface engineering for ultimate CMOS scaling*; 2021 Symposium on VLSI Technology, 13−19 June 2021; 2021; pp 1−2.

Recommended by ACS

Superior Cryotronics Performance of the In/p-WS_2 Schottky Interface

Ankit G. Dalvaniya, Vivek M. Pathak, *et al.*

MARCH 16, 2022
ACS APPLIED ELECTRONIC MATERIALS

READ

Statistical Assessment of High-Performance Scaled Double-Gate Transistors from Monolayer WS_2

Zheng Sun, Joerg Appenzeller, *et al.*

SEPTEMBER 12, 2022
ACS NANO

READ

Realization of Ultra-Scaled MoS_2 Vertical Diodes via Double-Side Electrodes Lamination

Wanying Li, Yuan Liu, *et al.*

MAY 26, 2022
NANO LETTERS

READ

Controlled Growth of Wafer-Scale Transition Metal Dichalcogenides with a Vertical Composition Gradient for Artificial Synapses with High Linearity

Lei Tang, Hui-Ming Cheng, *et al.*

AUGUST 01, 2022
ACS NANO

READ

Get More Suggestions >

https://doi.org/10.1021/acs.nanolett.2c02901
Nano Lett. 2022, 22, 7667−7673